

## Cryptography:

It is used to transform from an intelligence message into an unintelligence.

Plain text - original intelligence message

Cipher text - transformed message

### Cipher:

An algorithm for transforming by transposition and/or substitution methods.

### Key:

Information used by the cipher, known only to the sender & receiver.

### Encipher (encode)

Convert Plaintext to ciphertext using a cipher & key.

### Decipher (decode)

Convert ciphertext to Plain text using a cipher & key.

### Cryptography:

Classified into three independent dimensions.

i) Type of operations for transforming plaintext to ciphertext

- substitution
- transposition

ii) The num of keys used

- The sender & the receiver use same key  
symmetric key (or) single key encryption

- The sender and receiver use different keys  
Public key encryption

iii) The way in which the plain text is processed

- A block cipher
- A stream cipher

## Four types of attack

- 1) interruption - Attack on Availability
- 2) interception - Attack on integrity
- 3) Modification - Attack on authenticity
- 4) Fabrication - Attack on authenticity

## Passive Attack

- Implicit

Passive Attacks are two types

- 1) Release of message contents
- 2) Traffic analysis

## Active Attacks

There are four categories

- Masquerade
- Replay

## Exercise: Caesar cipher

Plain text: CYBER SECURITY

Solution

Step 1: shift each letter 3 positions backwards

Step 2: write the Plain Text

Recovered Plaintext:

CYBERSECURITY

C → F

T → W

Y → B

Y → B

B → E

E → H

R → U

S → V

E → H

C → F

U → X

R → U

I → L



## ii) Monalphabetic substitution cipher

Instead of shifting alphabets by fixed amount,

Ex:-

\* A is replaced by Q, B by D, C by T etc.

• The no. of alternative keys possible now becomes  $26!$ .

• Thus, Brute Force attack is impractical in this case.

• 'e' is the most commonly

• Encrypt the Plain text:

NETWORK

• Decrypt the ciphertext:

LTGXKOZEN

## iii) Playfair cipher

Step 1: Multiple characters (2 characters should convert)

Enter the key from left to right

Ignore repeated alphabets.

Step 2:

Divide the plain text into pair of letters

\* If two letters are in the same row, then replace with immediate right letters.

\* If two letters are in the same column, then replace with immediate below letters.

\* If the letters are not in the same row and column, draw a square as corners of PT, the remaining two corners becomes CT.

Plain text — good morning  $\Rightarrow$  go|od|mo|rn|in|g  
dz|voles|cu|rk|zg

|   |   |   |   |   |
|---|---|---|---|---|
| P | I | A | Y | F |
| H | N | E | X | M |
| B | C | D | G | L |
| K | Q | O | Z | S |
| T | U | V | W | R |

\* Encrypt & decrypt computer

Keyword: security  
Plaintext: computer.

compu|ter|con  
un|ni|ac|cs

|   |   |   |   |   |
|---|---|---|---|---|
| s | c | e | u | n |
| h | t | y | a | b |
| d | f | g | h | k |
| i | m | n | o | p |
| q | s | v | w | x |

Ciphertext: unniaccs.

① Scenario:

Event communication security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During the process, a message 'MEET AT NOON' is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

② Scenario: intercepted suspicious Message

A cybersecurity intern intercepts the coded message 'GSRH RH2HXNG' during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.



### Railfence:

Used to transposition the characters.  
The PT is written as a sequence of diagonals of any depth.  
Then read them as a sequence of rows, that becomes a CT.

Ex:-

PT - meet me after the class.

Diagonal depth: 2

|       |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|
| Row 1 | m | e | a | t | r | h | e |
| Row 2 | n | t | c | e | t | e | s |

C.T = mematrhcasetefetels.

Ex:-

key = laxor

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | a | s | e | r |
| 3 | l | x | o | r |

### Fiestal cipher structure

• It follows block cipher approach

• divided the text into different no. of blocks and perform operations on each block.

### Procedure

1) Divide the PT into 2 halves.

$n_i - 1$ , left half and  $r_i - 1$ , right half have the same structure.

2) perform some function that depends on the particular algorithm

Ex:- DES (or) RSA.

21/7/26

### Poly-Alphabetic Cipher:

- A set of monoalphabetic ciphers.
- If a letter is repeated more than once in plain text, then that is replaced by the same cipher text.
- All other letters are incremented by 1.

Ex- PT - Hello;  
CT = ifmmp

- In Polyalphabetic cipher, the same letter in plain text have different cipher text.

### Vigenere Cipher:

Let the sequence of PT,  $P = P_0 P_1 P_2 \dots P_{n-1}$

Key,  $K = K_0 K_1 K_2 \dots K_{m-1}, m \leq n$

CT,  $C = C_0 C_1 C_2 \dots C_{n-1}$

$$\begin{aligned} \therefore CT &= E(K, P) \\ &= E(K_0 K_1 K_2 \dots K_{m-1}) (P_0 P_1 P_2 \dots P_{n-1}) \\ &= (P_0 + K_0) \bmod 26, (P_1 + K_1) \bmod 26, \dots, (P_{m-1} + K_{m-1}) \bmod 26, \dots \end{aligned}$$

$$\begin{aligned} C_i &= (P_i + K_i \bmod m) \bmod 26 \\ P_i &= (C_i - K_i \bmod m) \bmod 26 \end{aligned}$$

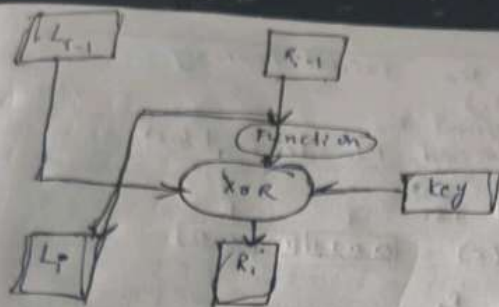
### Transposition Techniques:

substitution replace one letter with another letter in the plain text.

- key is some +1

Ex- P.T is abc  
C.T is bcd





- \* Each block size is 16 bytes, preferable.
- \* Key is 128 bit, preferable.

A Feistel cipher uses an 8-bit block and one round.  
 Plaintext = 10101101  
 Round key  $k_1 = 1100$   
 Round function :  $F(R, K) = R \oplus K$

Questions :

1) Split the Plaintext into  $L_0$  and  $R_0$

2) An 8-bit plaintext is encrypted into two Feistel rounds

Plaintext = 11100010

$k_1 = 0110$

$k_2 = 1001$

3) A ciphertext 00111100 was produced by a one-round Feistel cipher

Round key  $k_1 = 0101$

$F(R, K) = R \oplus K$

Questions :

1) Split the ciphertext into  $L_1$  and  $R_1$ .

- 1) Recover  $R_0$  using the Feistel property
- 2) Compute  $F(R_0, K_1)$
- 3) Recover the original Plaintext block
- 4) Write the recovered Plaintext block

Ans) Given cipher text is

Cipher text (C) =  $\begin{matrix} L & R \\ 0011 & 1100 \end{matrix}$

1) Splitting  $L_1$  &  $R_1$

$$\begin{matrix} L_1 = 0011 \\ R_1 = 1100 \end{matrix}$$

2) Recover  $R_0$

In Feistel Property

$$R_0 = L_1$$

$$\therefore R_0 = 0011$$

3) Compute  $F(R_0, K_1)$

Given function

$$F(R, K) = R \oplus K$$

Given

$$K_1 = 0101$$

$$\therefore F(R_0, K_1) = R_0 \oplus K_1$$

| $R_0$ | $K_1$ | $R_0 \oplus K_1$ |
|-------|-------|------------------|
| 0     | 0     | 0                |
| 0     | 1     | 1                |
| 1     | 0     | 1                |
| 1     | 1     | 0                |

$$\therefore F(R_0, K_1) = 0110$$



### Hill cipher

- Hill cipher is a classical polyalphabetic substitution cipher.
- It uses matrix multiplication to encrypt blocks of letters.

#### steps

- Convert letters into numbers.
- Group letters into blocks (usually 2 (or) 3 letters).
- multiply

The general equation of cipher text is

$$C = KP \pmod{26}$$

$$(C_1 C_2) = \begin{pmatrix} K_1 & K_2 \\ K_3 & K_4 \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \pmod{26}$$

#### Decryption

$$P = K^{-1} \times C \pmod{26}$$

#### Letter to Number Mapping

|   |    |
|---|----|
| A | 0  |
| B | 1  |
| C | 2  |
| ⋮ | ⋮  |
| Z | 25 |

$$K = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

$$P = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$$

Ex:-

Hello:

|   |   |    |
|---|---|----|
| H | - | 7  |
| e | - | 4  |
| l | - | 11 |
| l | - | 11 |
| o | - | 14 |
| X | - | 23 |

$$(C_1 C_2) = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{matrix} 14 + 12 \\ 28 + 20 \end{matrix}$$

$$(C_1 C_2) = \begin{pmatrix} 26 \\ 48 \end{pmatrix} \quad \begin{matrix} 26 \pmod{26} = 0 \\ 48 \pmod{26} = 22 \end{matrix}$$

$$0 = A$$

$$22 = W$$

4) Recover  $h_0$

Formula

$$h_0 = R_1 \oplus F(R_0, k_1)$$

We know that

$$R_1 = 1100$$

$$F(R_0, k_1) = 0110$$

| $R_1$ | $F(R_0, k_1)$ | $R_1 \oplus F(R_0, k_1)$ |
|-------|---------------|--------------------------|
| 1     | 0             | 1                        |
| 1     | 1             | 0                        |
| 0     | 1             | 1                        |
| 0     | 0             | 0                        |

$$\therefore h_0 = 1010$$

5) Recovered Plaintext block

$$h_0 = 1010$$

$$R_0 = 0011$$

$$h_0 || R_0$$

Plaintext : 10100011

Encryption

$$C = E(K, P)$$

$$C = (P + K) \bmod 26$$

Decryption

$$P = D(K, C)$$

$$P = (C - K) \bmod 26$$



$$K = \begin{pmatrix} 2 & 2 \\ 4 & 4 \end{pmatrix} \quad P = \begin{pmatrix} 11 \\ 11 \end{pmatrix}$$

$$(C_1, C_2) = \begin{pmatrix} 2 & 2 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix} \pmod{26} = \begin{pmatrix} 22 + 22 \\ 44 + 44 \end{pmatrix} \pmod{26} = \begin{pmatrix} 44 \pmod{26} = 18 \\ 88 \pmod{26} = 21 \end{pmatrix}$$

$$\text{Cipher numbers} = (18, 21)$$

$$18 = D$$

$$21 = V$$

OX

$$(C_1, C_2) = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 23 \end{pmatrix} \pmod{26} = \begin{pmatrix} 47 \\ 171 \end{pmatrix} \pmod{26} = \begin{pmatrix} 19 \\ 15 \end{pmatrix}$$

$$C.N = (19, 15)$$

$$19 = T$$

$$15 = P$$

$$\therefore \text{Cipher Text} = AWDVTP$$

Decryption:-

$$P = K^{-1} \times C \pmod{26}$$

$$K = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

$$|K| = (2 \times 5) - (3 \times 4) = 10 - 12 = -2$$

$$\therefore 26 - 2 = 24$$

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

$$H = 7$$

$$E = 4$$

$$L = 11$$

$$L = 11$$

$$O = 14$$

$$X = 23$$

HE

Encryption

$$(c_1, c_2) = \begin{pmatrix} k_1 & k_2 \\ k_3 & k_4 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \mod 26$$

$$(c_1, c_2) = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} \mod 26$$

$$33 \mod 26 = 7$$

$$34 \mod 26 = 8$$

$$(c_1, c_2) = (7, 8) = (H, I)$$

LL

$$(c_1, c_2) = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix} \mod 26$$

$$= 66 \mod 26 = 14$$

$$77 \mod 26 = 25$$

$$(c_1, c_2) = (14, 25) = (O, Z)$$

OX

$$(c_1, c_2) = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 23 \end{pmatrix} \mod 26$$

$$111 \mod 26 = 7$$

$$143 \mod 26 = 13$$

$$(c_1, c_2) = (7, 13) = (H, N)$$

Cipher Text = H I O Z H N



Block 1

$$P_1 = (15 \times 7 + 17 \times 8) \bmod 26$$

$$= 241 \bmod 26 = 7$$

$$P_2 = (20 \times 7 + 9 \times 8) \bmod 26$$

$$= 212 \bmod 26 = 4$$

$7 \rightarrow H$     $4 \rightarrow E$

Block 2

$$P_1 = (15 \times 14 + 17 \times 25) \bmod 26 = 635 \bmod 26 = 11$$

$$P_2 = (20 \times 14 + 9 \times 25) \bmod 26 = 505 \bmod 26 = 11$$

$11 \rightarrow L$     $11 \rightarrow L$

Block 3

$$P_1 = (15 \times 7 + 17 \times 13) \bmod 26 = 326 \bmod 26 = 14$$

$$P_2 = (20 \times 7 + 9 \times 13) \bmod 26 = 257 \bmod 26 = 23$$

$14 \rightarrow O$     $23 \rightarrow X$

$\therefore$  HELLO

25/7/26 :-

~~$k_i = R_i - 1$~~

Unit

$$\begin{bmatrix} 11 & 21 \\ 17 & 09 \end{bmatrix} = \begin{bmatrix} 11 & 21 \\ 17 & 09 \end{bmatrix}$$

Decryption:-

$$P = K^{-1} \times C \pmod{26}$$

$$K = \begin{pmatrix} 5 & 3 \\ 2 & 3 \end{pmatrix} \quad ad-bc$$

$$K = 15 - 6$$

$$K = 9$$

$$C = K \times P \pmod{26}$$

Inverse of 9 modulo 26 is 3

$$9 \times 3 = 27 \equiv 1 \pmod{26}$$

Adjoint of K

Formula

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & -3 \\ -2 & 5 \end{bmatrix}$$

convert negative values

$$26 - 3 = 23 \pmod{26}$$

$$26 - 2 = 24 \pmod{26}$$

$$-3 \rightarrow 23$$

$$-2 \rightarrow 24$$

$$9 \times 3 = 27 \equiv 1$$

$$27 - 26 = 1$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & 23 \\ 24 & 5 \end{bmatrix}$$

multiply by 3

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

Decrypt Cipher

$$\text{Cipher} = \text{HIOZHN}$$

$$H = 7$$

$$I = 8$$

$$O = 14$$

$$Z = 25$$

$$H = 7$$

$$N = 13$$



25/7/22

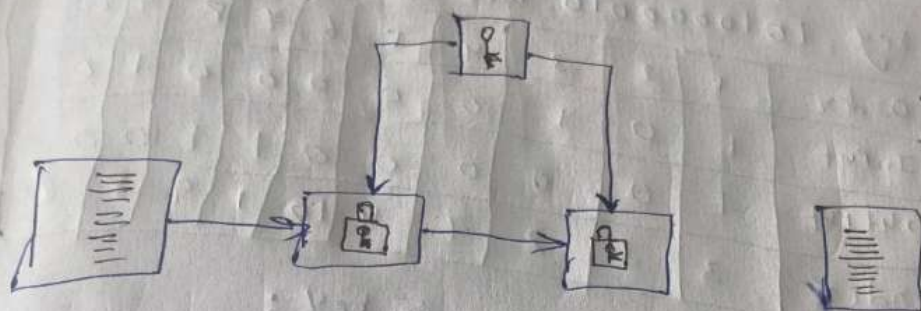
## Unit - II :-

### Symmetric Key Cryptography :-

A symmetric key cryptosystem / secret-key cryptography / private-key cryptography.

Techniques used:

- Substitution
- Transposition



### Types of symmetric key cryptography:

- 1) Block cipher  $\rightarrow$  Encrypts fixed-size blocks
- 2) stream cipher  $\rightarrow$  Encrypts one bit or one byte at a time.

### Simplified Data Encryption Standard (S-DES)

S-DES is a simple version of the DES Algorithm.

It is a block cipher.

It is a symmetric key cipher

S-DES is use 8 bits in Encryption  
Plain-text.

Accept the 10-bit key & convert it into 8-bit

step-5:

performing 10-bit convert to 8-bit

| order         | 1 | 2 | 3 | 4 | 5 | 6 | 7  | 8 | 9 | 10 |
|---------------|---|---|---|---|---|---|----|---|---|----|
| combined bits | 0 | 0 | 0 | 0 | 1 | 1 | 0  | 0 | 0 | 0  |
| output order  | 6 | 5 | 7 | 4 | 8 | 5 | 10 | 9 |   |    |
| output        | 1 | 0 | 1 | 0 | 0 | 1 | 0  | 0 |   |    |

$K_1$  is = 10100100.

step-6:

we require two keys S-D-E has two create second key

copy both of the halves in step 4. i.e., the results of the first round shift

skip the unused bits

1st half : 00001; 2nd half : 11000

step-7:

Performing two-bit round shift.

After two-bit round shift: 1st half : 00100,

2nd half : 00011

combining : 001000011

step-8:

Perform Permutation P

| order        | 1 | 2 | 3 | 4 | 5 | 6 | 7  | 8 | 9 | 10 |
|--------------|---|---|---|---|---|---|----|---|---|----|
| Input        | 0 | 0 | 1 | 0 | 0 | 0 | 0  | 0 | 1 | 1  |
| output order | 6 | 3 | 7 | 4 | 8 | 5 | 10 | 9 |   |    |
| output       | 0 | 1 | 0 | 0 | 0 | 0 | 1  | 1 |   |    |

$K_2$  : 01000011



key = 1010000010  
 Plain text = 11100001

step-2:

Then the key should proceed with permutation 10.

P10 table

|              |   |   |   |   |   |    |   |   |   |    |
|--------------|---|---|---|---|---|----|---|---|---|----|
| Input order  | 1 | 2 | 3 | 4 | 5 | 6  | 7 | 8 | 9 | 10 |
| Output order | 3 | 5 | 2 | 7 | 4 | 10 | 1 | 9 | 8 | 6  |

put the key 1010000010 into the P10 table:

|        |   |   |   |   |   |    |   |   |   |    |
|--------|---|---|---|---|---|----|---|---|---|----|
| Order  | 1 | 2 | 3 | 4 | 5 | 6  | 7 | 8 | 9 | 10 |
| Input  | 1 | 0 | 1 | 0 | 0 | 0  | 0 | 0 | 1 | 0  |
| Output | 1 | 0 | 0 | 0 | 0 | 0  | 1 | 1 | 0 | 0  |
| Order  | 3 | 5 | 2 | 7 | 4 | 10 | 1 | 9 | 8 | 6  |

output of the key after P10 is key:

output : 1 0 0 0 0 0 1 1 0 0

step-3:

Dividing the key

1st half : 10000

2nd half : 01100

step-4:

Apply one-bit round shift (left shift) on each half.

Before one-bit round shift : 1st half :

10000 → 00001

After 1

00001

11000

Combine both - 000011000



ii) Encryption of Plain text into cipher text in 8 bits.  
 The input & output is 8 bits. or  
 the block size is always 8 bits  
 Plain Text : 0110010

Step-1:  
 performing IP-8, initial permutation on plain text  
 output bits : 10101001

|              | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|--------------|---|---|---|---|---|---|---|---|
| order        |   |   |   |   |   |   |   |   |
| Input Bits   | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| permute bits | 2 | 6 | 3 | 1 | 4 | 8 | 5 | 7 |
| output       | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |

Step-2:-

Dividing

1<sup>st</sup> : 1010      2<sup>nd</sup> : 1001

Step-3: Take right half & expand it.

|             |   |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|---|
| Right half  | 1 | 0 | 0 | 1 |   |   |   |   |
| order       | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Expand Bits | 4 | 1 | 2 | 3 | 2 | 3 | 4 | 1 |
| output      | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 |

Step-4:

Take the output and XOR with the First key K<sub>1</sub>

Key 1 : 10000000 (step 5, ~~key~~)

output Bits : 11000011 (step 3, P.T)

Key-1 XOR bits : (10100100) XOR (11000011)

Output : 01100111



Now put each half into the S-boxes, two choices: S-0 and S-1.

S-0

For left Half: 0110

Row: 00 = 0 (1st row)

col: 11 = 3 (3rd col)

So, 2 = 10.

|   |   |   |   |
|---|---|---|---|
| S | 0 |   |   |
| 1 | 0 | 3 | 2 |
| 3 | 2 | 1 | 0 |
| 0 | 2 | 1 | 3 |
| 3 | 1 | 3 | 2 |

|   |   |   |   |
|---|---|---|---|
| S | 1 |   |   |
| 0 | 1 | 2 | 3 |
| 2 | 0 | 1 | 2 |
| 3 | 0 | 1 | 0 |
| 2 | 1 | 0 | 3 |

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 2 |
| 1 | 1 | 3 |

S-1

For Right Half: 0111

Row: 01 = 1 (1st row)

col: 11 = 3 (3rd col)

Combining the two halves: 1011

So S1 = 3 = 11

Step-6:

Performing Permutation 4:

Output: 0111

| order        | 1 | 2 | 3 | 4 |
|--------------|---|---|---|---|
| Input        | 1 | 0 | 1 |   |
| output order | 2 | 4 | 3 | 1 |
| output       | 0 | 1 | 1 | 1 |

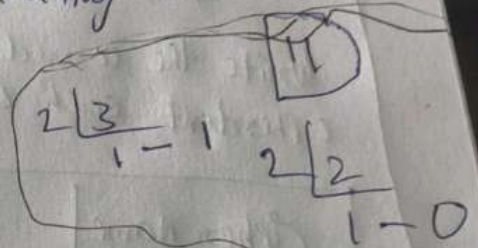
Step-7:

XOR this output with the final 4 bits of initial Permutation.

Step 2 contains the initial Per-remaining mutation bit's

0111 XOR 1010 = 1101

Output: 1101



Step-8:

combine the output with right half of the initial Permutation which is in step 2.

8-bit: 11011001

Step-9:

Dividing the two equal halves & swapping them

Left: 1001 Right halves: 1101

After swapping



step-10?

10011101

### Exercise-1

match the cipher name with the Encryption Rule:  
 Caesar cipher - shifts every letter by a fixed number  
 Vigenere cipher - using repeating keyword with multiple shifts  
 Rail Fence cipher - Repeating

### Exercise-2:

- |       |       |
|-------|-------|
| A -   | A - 3 |
| B -   | B - 2 |
| C -   | C - 6 |
| D - 4 | D - 5 |
| E - 6 | E - 1 |
| F - 5 | F - 4 |
| F -   |       |

Write the correct null, advantage, weakness and cipher text each cipher:

| Cipher Name                     | Encryption Rule                                                    | Advantages                                 | Weakness                                                | Sample Cipher Text   |
|---------------------------------|--------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------|----------------------|
| Caesar cipher                   | $E = E(k, P)$<br>$C = (P+k) \bmod 26$                              | • Very simple & easy to                    | Easy to break using brute force (only 25 keys)          | HELLO → KHOOR (K=3)  |
| Monoidal Algebraic Substitution | $C = f(P)$                                                         | large key space than Caesar cipher         | vulnerable to frequency analysis                        | HELLO → ITSS         |
| Vigenere cipher                 | $C_i = (P_i + k_i) \bmod 26$                                       | more secure than Caesar w/ multiple shifts | Can be broken if the key length is discovered           | ATTACK → LEH (key)   |
| Rail Fence cipher               | write Plain text in zig-zag pattern across rails and read row-wise | Easy to implement, hides letter positions  | Not secure against modern attacks                       | HELLO → LXFOL        |
| Columnar Transposition          | write Plain text in rows under key word                            | preserve letter frequencies                | can be broken with enough cipher text                   | HELLO → FLHLL (name) |
| Hill cipher                     | $C = K \times P \bmod 26$                                          | Encrypt blocks of letters                  | Requiring inverse matrix, vulnerable to known plaintext | HELLO → HJAT         |